

PI-52-2018

1. Considere la siguiente igualdad:

$$1 \cdot 5^1 + 2 \cdot 5^2 + 3 \cdot 5^3 + \dots + n \cdot 5^n = \frac{5 + (4n-1) \cdot 5^{n+1}}{16}$$

a) Demuestre, usando inducción matemática que dicha igualdad es válida para toda  $n \geq 1$ .

$$n=1 \quad 1 \cdot 5^1 = \frac{5 + (4 \cdot 1 - 1) \cdot 5^{1+1}}{16}$$

$$5 = 5 \quad \checkmark$$

$$n=p \quad 1 \cdot 5^1 + 2 \cdot 5^2 + 3 \cdot 5^3 + \dots + p \cdot 5^p = \frac{5 + (4p-1) \cdot 5^{p+1}}{16}, \text{ H.I.}$$

$$n=p+1 \quad 1 \cdot 5^1 + 2 \cdot 5^2 + 3 \cdot 5^3 + \dots + p \cdot 5^p + (p+1) \cdot 5^{p+1} =$$
$$\frac{5 + (4(p+1)-1) \cdot 5^{(p+1)+1}}{16} \quad \text{H.O.D.}$$

$$\frac{5 + (4p+4-1) \cdot 5^{p+2}}{16}$$

$$\frac{5 + (4p+3) \cdot 5^{p+2}}{16}$$

$$\frac{5 + (4p+3) \cdot 5 - 5^{p+1}}{16}$$

$$\frac{5 + (20p+15) \cdot 5^{p+1}}{16}$$

Demostracion

$$1 \cdot s^1 + 2 \cdot s^2 + 3 \cdot s^3 + \dots + p \cdot s^p + (p+1) \cdot s^{p+1}$$

$$\frac{s + (7p-1) \cdot s^{p+1}}{16} + (p+1) \cdot s^{p+1} \quad \text{H.I.}$$

$$\frac{s + (7p-1) \cdot s^{p+1}}{16} + (p+1) \cdot s^{p+1} \quad , \frac{16}{16}$$

$$\frac{s + (7p-1) \cdot s^{p+1}}{16} + \frac{(p+1) \cdot s^{p+1} \cdot 16}{16}$$

$$\frac{s + (7p-1) \cdot s^{p+1} + (p+1) \cdot s^{p+1} \cdot 16}{16}$$

$$\frac{s + s^{p+1} (7p-1 + 16(p+1))}{16}$$

$$\frac{s + s^{p+1} (7p-1 + 16p + 16)}{16}$$

$$\frac{s + s^{p+1} (20p + 15)}{16}$$

$$\frac{s + s^{p+1} \cdot 5(7p+3)}{16}$$

$$\boxed{\frac{s + (7p+3) \cdot s^{p+2}}{16}}$$

$$\sum_{n=p}^{\infty} a_n = b_n$$

Veremos si  
 $b_n$  converge, NO  
 la suma

b) Si  $\{S_n\} = \left\{ \frac{5 + (4n-1) \cdot 5^{n+1}}{16} \right\}$  corresponde a la sucesión de sumas parciales asociada

$$\infty \cdot \infty = \infty$$

a la serie  $\sum_{k=1}^{\infty} k \cdot 5^k$ , determine si dicha serie converge o diverge.

(2 puntos)

$$\lim_{n \rightarrow \infty} \frac{5 + (4n-1) \cdot 5^{n+1}}{16} = \frac{5 + (4 \cdot \infty - 1) \cdot 5^{+\infty+1}}{16}$$

$$= \frac{5 + \infty \cdot \infty}{16} = \infty$$

Diverge

2. Determine todos los valores de  $b$  para que de la serie  $\sum_{n=5}^{\infty} \left( \frac{n}{b^n} - \frac{n+1}{b^{n+1}} \right)$  converja y su suma

$$\lim_{n \rightarrow \infty} \frac{n}{b^n} - \frac{n+1}{b^{n+1}}$$

geométrica  
 Telescópica

$$\frac{5}{5^5} - \lim_{n \rightarrow \infty} \frac{n+1}{b^{n+1}} \approx \frac{n}{b^n} = 0$$

$$\frac{5}{5^5} \quad b > 5$$

3. Determine el intervalo de convergencia (NO incluya el análisis de los extremos) de la serie siguiente. (4 puntos)

$$\sum_{n=1}^{\infty} \frac{4^n \cdot (n+2)! (x-3)^n}{5 \cdot 8 \cdot 11 \cdots (3n+2)}$$

Serie de potencias  
↳ Razon

$$\lim_{n \rightarrow +\infty} \frac{4^{n+1} \cdot (n+3)! \cdot (x-3)^{n+1}}{5 \cdot 8 \cdot 11 \cdots (3n+2)(3n+5)} \cdot \frac{5 \cdot 8 \cdot 11 \cdots (3n+2)}{4^n \cdot (n+2)! \cdot (x-3)^n}$$

$3(n+1)+2$   
 $3n+3+2$   
 $3n+5$

$$\lim_{n \rightarrow +\infty} \frac{4^{n+1} \cdot (n+3)! \cdot (x-3)^{n+1}}{4^n \cdot (n+2)! \cdot (x-3)^n} \cdot \frac{5 \cdot 8 \cdot 11 \cdots (3n+2)}{5 \cdot 8 \cdot 11 \cdots (3n+2)(3n+5)}$$

$$n! = n(n-1)(n-2) \cdots 1$$

$$\lim_{n \rightarrow +\infty} \frac{4^{n+1} \cdot (n+3)! \cdot (x-3)^{n+1}}{4^n \cdot (n+2)! \cdot (x-3)^n (3n+5)}$$

$$\lim_{n \rightarrow +\infty} \frac{4^{n+1} \cdot (n+3)(n+2)! \cdot (x-3)^{n+1}}{4^n \cdot (n+2)! \cdot (x-3)^n (3n+5)}$$

$\rightarrow (n+3)! = (n+3)(n+2)! = (n+3)(n+2)!$

$$|x-3| \lim_{n \rightarrow +\infty} \frac{4^{n+1}}{3n+5}$$

$$L = K |x-c|$$

$$|x-3| \lim_{n \rightarrow +\infty} \frac{4^n}{3n} = \frac{4}{3}$$

Condición de convergencia

$$K |x-c| < 1$$

$$|x-3| < \frac{1}{K}$$

$$-\frac{2}{K} < x-c < \frac{2}{K}$$

$$-\frac{2}{K} + c < x < \frac{2}{K} + c$$

$$\frac{4}{3} \cdot |x-3| < 1$$

$$4 |x-3| < 3 \quad \rightarrow \quad -\frac{3}{4} < x-3 < \frac{3}{4}$$

$$|x-3| < \frac{3}{4} \quad \rightarrow \quad -\frac{3}{4} + 3 < x < \frac{3}{4} + 3$$

$$\frac{9}{4} < x < \frac{15}{4}$$

$$I = x \in \left] \frac{9}{4}, \frac{15}{4} \right]$$

$$-\frac{\pi}{2} < \arctan(n) < \frac{\pi}{2}$$

4. Determine si cada una de las siguientes series convergen o divergen. Debe indicar los criterios aplicados en cada caso.

a)  $\sum_{k=5}^{\infty} \frac{\arctan k + 4}{(k-4)^2}$

TODAS Con trigonometricos

Se usa comparacion directa (4 puntos)

$$-\frac{\pi}{2} < \arctan(k) < \frac{\pi}{2}$$

$$-\frac{\pi}{2} + 4 < \arctan(k) + 4 < \frac{\pi}{2} + 4$$

$$\frac{-\frac{\pi}{2} + 4 < \arctan(k) + 4 < \frac{\pi}{2} + 4}{(k-4)^2} \quad \frac{-\frac{\pi}{2} + 4 < \arctan(k) + 4 < \frac{\pi}{2} + 4}{(k-4)^2}$$

Elegir Cualquiera

$\infty$

$$\sum_{k=5}^{\infty} \frac{\frac{\pi}{2} + 4}{(k-4)^2}$$

$$k=5 \quad (k-4)^2$$

$\infty$

$$\frac{\frac{\pi}{2} + 4}{(k-4)^2} \leq \frac{1}{(k-4)^2}$$

$\infty$

$$\frac{\frac{\pi}{2} + 4}{(k-4)^2} \leq \frac{1}{k^2} \quad \text{p serie } p > 1$$

$\therefore$  Converge

$$\therefore \frac{\arctan(k) + 4}{(k-4)^2} \text{ Converge}$$

NO estan diciendo que es  
alternada, NO usar ese criterio

$$b) \sum_{n=1}^{\infty} \frac{(-1)^n \cdot \left(2 + \frac{1}{n}\right)^{2n}}{e^n}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} \quad \begin{array}{l} \text{Valor} \\ \text{ABSOLUTO} \end{array}$$

$$\lim_{n \rightarrow \infty} \frac{\sqrt[n]{1-1} \cdot \sqrt[n]{\left(2 + \frac{1}{n}\right)^{2n}}}{\sqrt[n]{e^n}} \quad x^{2n} = (x^2)^n$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\left(2 + \frac{1}{n}\right)^2}{e} &= \frac{\left(2 + \frac{1}{\infty}\right)^2}{e} \\ &= \frac{(2+0)^2}{e} \\ &= \frac{4}{e} > 1 \end{aligned}$$

∴ Diverge

$$c) \sum_{n=1}^{\infty} \left[ \frac{(-1)^{3n} \cdot n^2}{n^3 + 1} - \frac{1}{3^{-n+4}} \right]$$

Si bn diverge, todo diverge  
Si an diverge, Todo diverge

$$\sum_{n=1}^{\infty} \frac{(-1)^{3n} \cdot n^2}{n^3 + 1} - \sum_{n=1}^{\infty} \frac{1}{3^{-n+4}}$$

↳

↳ Geométrica

$$\sum_{n=1}^{\infty} \frac{1}{3^{-n} \cdot 3^4} = \frac{1}{3^4} \sum_{n=1}^{\infty} \frac{1}{3^{-n}} = \frac{1}{3^4} \sum_{n=1}^{\infty} 3^n = 13 \cdot 3 > 1$$

∴ Diverge

∴ Todo diverge

5. Si la función  $f : [2, +\infty[ \rightarrow \mathbb{R}$  tal que  $f(x) = \frac{1}{x(\ln x)^2}$  es continua, positiva y decreciente, determine si la serie  $\sum_{k=2}^{\infty} (-1)^{k+1} \cdot \frac{1}{k(\ln k)^2}$  diverge, converge absolutamente o converge condicionalmente. (4 puntos)

$\infty$

$$\sum_{k=2}^{\infty} (-1)^{k+1} \cdot \frac{1}{k(\ln k)^2}$$

Por crit de integral

NO se hace la parte de verificar si es decreciente por el ENUNCIADO!!

$\infty$

$$\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$$

$$\int_2^{+\infty} \frac{1}{x(\ln x)^2} dx \rightarrow \int \frac{1}{x(\ln x)^2} dx \quad u = \ln(x) \quad du = \frac{1}{x} dx$$

$$\int \frac{1}{u^2} du$$

$$\int u^{-2} = \frac{u^{-2+1}}{-2+1} = \frac{u^{-1}}{-1}$$

$$= -\frac{1}{u}$$

$$\rightarrow \left. -\frac{1}{\ln(x)} \right|_2^{+\infty} = -\frac{1}{\ln(k)}$$

$$\lim_{k \rightarrow \infty} \frac{-1}{\ln(k)} - \left( -\frac{1}{\ln(2)} \right) = \frac{1}{\ln(2)} \therefore \text{Converge}$$

$\therefore$  Converge Absolutamente

6. Asuma que la serie  $S = \sum_{k=1}^{\infty} \frac{(-1)^k \cdot 4^k}{(2k+1) \cdot (2k+1)!}$  es convergente. Determine el menor valor para  $n$  de manera que  $S_n$  aproxime a  $S$  con un error menor a  $10^{-6}$ . (3 puntos)

$$N \text{ (results)} \quad N+1 \text{ (Para probar)} \quad a_n < 10^{-6} \\ < 0,000001$$

3

7

$$\frac{7^7}{(2 \cdot 7 + 1)(2 \cdot 7 + 1)!}$$

$$= 0,00007 \not< 0,000001$$

5

6

$$\frac{7^6}{(2 \cdot 6 + 1)(2 \cdot 6 + 1)!}$$

$$0,00000005 < 0,000001$$

$$\boxed{R/ \quad N=5}$$

PI - SI - 2025

#1 Considere la sucesión  $\{a_n\}_{n \geq 1}$  definida por:

$$a_n = \frac{2^n \cdot n}{n!}$$

Determine si la sucesión es creciente o decreciente a partir de algún  $n$ .

$$\frac{a_{n+1}}{a_n} \geq 1 \quad V \rightarrow \text{Crecer}$$

$$F \rightarrow \text{Decrecer}$$

$$\frac{2^{n+1} \cdot (n+1)}{(n+1)!} \geq 1$$
$$\frac{2^n \cdot n}{n!}$$

$$\frac{2^{n+1} \cdot (n+1) n!}{2^n \cdot n (n+1)!} \geq 1$$

$$\frac{\cancel{2} \cdot \cancel{2}^1 \cancel{(n+1)} \cdot \cancel{n!}}{\cancel{2}^1 \cdot n \cdot \cancel{(n+1)} \cdot \cancel{n!}} \geq 1$$

$$\frac{2}{n} \geq 1$$

$$n \geq 2 \quad \text{Crecer}$$

$$n \leq 2 \quad \text{Decrecer}$$

$$2 \geq n$$

$$n \leq 2$$

si Decrece

#2 Sea  $S = \sum_{j=1}^{\infty} (3j-2)$ . Pruebe por inducción que la  $N$ -ésima suma parcial de  $S$  es igual a (4 Pts)

$$S_N = \frac{N(3N-1)}{2}, \text{ para todo entero } N \geq 1 \quad n = N$$

$$n=1 \quad 3 \cdot 1 - 2 = \frac{1(3 \cdot 1 - 1)}{2}$$

$$1=1$$

$$n=p \quad \begin{matrix} p \\ \in \\ n=1 \end{matrix} (3j-2) = \frac{p(3p-1)}{2}, \text{ H.I.}$$

$$n=p+1 \quad \begin{matrix} p+1 \\ \in \\ n=1 \end{matrix} (3j-2) = \frac{(p+1)(3p+2)}{2}, \text{ H.Q.D.}$$

$\begin{matrix} 3(p+1)-1 \\ 3p+3-1 \\ = 3p+2 \end{matrix}$

$$= \frac{3p^2 + 2p + 3p + 2}{2}$$

$$= \frac{3p^2 + 5p + 2}{2}$$

Demonstracion

$$\begin{matrix} p+1 \\ \in \\ n=1 \end{matrix} (3j-2)$$

$$\begin{matrix} p \\ \in \\ n=1 \end{matrix} (3j-2) + (3p+1)$$

$\begin{matrix} 3(p+1)-2 \\ 3p+3-2 \\ = 3p+1 \end{matrix}$

$$\begin{matrix} p \\ \in \\ n=1 \end{matrix} (3j-2) + (3p+1)$$

$$\frac{p(3p-1)}{2} + (3p+1), \text{ H.I.}$$

$$\frac{p(3p-1)}{2} + (3p+1) \cdot \frac{2}{2}$$

$$\frac{p(3p-1) + 2(3p+1)}{2}$$

$$\frac{3p^2 - p + 6p + 2}{2}$$

$$\frac{3p^2 + 5p + 2}{2}$$

$$\boxed{\frac{(p+1)(3p+2)}{2}}$$

#3 Calcule la suma de las siguientes series

$$a) \sum_{k=2}^{\infty} \frac{(-3)^{k+1}}{4^k}$$

$\infty$

$$\sum_{k=2}^{\infty} \frac{(-3)^k \cdot (-3)^1}{4^k}$$

$$k=2$$

$$-3 \cdot \sum_{k=2}^{\infty} \left( \frac{-3}{4} \right)^k \cdot \frac{r^p}{1-r}$$

$$-3 \cdot \frac{\left( \frac{-3}{4} \right)^2}{1 - \frac{-3}{4}} = \boxed{\frac{-27}{28}}$$

#3 Calcule la suma de las siguientes series

$$b) \sum_{n=3}^{\infty} \frac{5n - (n+1)}{5^{n+1}}$$

$\infty$

$$\sum_{n=3}^{\infty} \frac{5n}{5^{n+1}} - \frac{n+1}{5^{n+1}}$$

$\infty$

$$\sum_{n=3}^{\infty} \frac{\cancel{5}n}{5^n \cdot \cancel{5}} - \frac{n+1}{5^{n+1}}$$

$\infty$

$$a_n - a_{n+1}$$

$$\sum_{n=3}^{\infty} \frac{n}{5^n} - \frac{n+1}{5^{n+1}}$$

$$\frac{3}{5^3} - \lim_{n \rightarrow \infty} \frac{n+1}{5^{n+1}} \quad n < a^n$$

$$\boxed{\frac{3}{125}}$$

$$\frac{f(x)}{g(x)}, \quad g(x) > f(x) \\ = 0$$